

Course Project Milestone3

Automated Warehouse Scenario

Yuh-Chen Liu

Arizona State University
yliu927@asu.edu

Abstract

The Automatic Warehouse Scenario is a real live problem that we would like to solve. The purpose of this question is to minimize the expense of warehousing by optimize the robotics movement. This report describes the issue we are facing and our approach to solve the issue.

Problem Statement

Automatic Warehouse Scenario describes a situation for several robots located in a limited space rectangular grid warehouse to deliver specific products on the shelves to the picking-stations by specific orders.

The robots are designed to be flat so they can go underneath the shelves and lift the shelves and move them to the designated picking station. The robots are designed to have the following movements, and at any state of these movements, collision of two robots must be avoided:

- Move horizontally or vertically to adjacent cells.
- Pick up and put down shelves.
- Deliver products or Idle.

There are also the following objects with its own definition or constraints that warehouse is consist of:

- A *node* is the basic unit of the warehouse, every object that is in the warehouse is on top of a node.
- A *highway* is a unit that no shelves can be placed on.
- A *picking station* is a unit where robots deliver the products on the shelves to.
- A *shelf* is a unit where is used to stock the products waiting to be delivered.
- A *product* is located on a shelf waiting for delivery.

The goal is to generate a schedule of robot movements that minimize the time cost to complete the orders.

Progress Made

To complete this project, we have broken down the tasks to the following segments: **The movement of the robots, the handling and re-routing for robots to pick up, put**

down, move around with the shelves, the identification of the location of the product and the picking station, and lastly, the handle of the orders. For now, we have completed the first segment, to handle the expecting movement of the robots.

Issues or Challenges Encountered

Currently we are facing the issue to handle the status of the shelves being lifted. This is because we have not yet established the correct constraints for shelf movements. The bottleneck of this issue is because the movement of the shelf is in sync with the robot picking it up. We also need to account for the movement of a robot after a shelf has been lift by the robot.

Issues Resolving Plan

We further discuss our plan to handle shelves movement issue. We therefore introduce a new state, $on(R,S,T)$, as the representation of a shelf S is on a robot R at time T . This state follows the commonsense law of inertia until put-down is executed. We can further examine the status of a shelf with the robot that picks it up. Below are the test cases we setup to achieve our goal of moving the shelves:

- $\therefore not\ pickup(1,1,m)$. This goal enforce *robot 1* to pick up *shelf 1* and time m . Which gives us the information of whether the robot pick up the shelf as expected, and where the robot picks up the shelf.
- $\therefore not\ on(1,1,m)$. In this case, we then try to derive the fact that the *shelf 1* is on the *robot 1*, and it stays that way at time m .
- $\therefore not\ loc(shelf,1,1,m)$. In this case, we would proceed to start trying to move shelf, at this point, we just need to shelf to be moved to designated position, which is *shelf 1, at(1,1)*, at time m , the constraints of highway, or not being able to move across shelves are not in scope.

Working Items

Completed Working Items

So far, we have read through the requirements and samples of the project stated in *ASP challenge 2019, ASP Challenge Problem: Automated Warehouse Scenario.*, and we have broken down the project into 4 big segments. Among these 4 benchmarks, we have completed the first portion. And we further dissect the second segment into smaller tasks that we can manage, among these tasks, we have completed the following:

- **The movement of the robots and constraints**
- **The handling and re-routing for robots to pick up, put down, move around with the shelves.** Below are the completed tasks in this partially done segment:
 - The exogeny of the *pickup, putdown* actions.
 - The uniqueness and existence of value constraints for shelves fluent.
 - The concept of $on(R,S,T)$ to express the state of a shelf being lift.
 - The commonsense law of inertia for $on(R,S,T)$ and $loc(shelf,N,X,Y,T)$.
 - The Direct effect of shelf movement.

Future Working Item and Estimated Schedule

There are still two larger working items needed to be breakdown and one larger working item is still half-done, here we will list out the remaining working item:

- **The identification of the product location and the picking station.** Need to breakdown the segment to smaller tasks, estimated workload, 2 days.
- **The handle of orders.** Need to breakdown the segment to smaller tasks, estimated workload, 3 days.
- **The handling and re-routing for robots to pick up, put down, move around with the shelves.** Below we will list out the undone working items and the estimated time:
 - The constraints of robot movement when picking up shelf. Estimated workload, 2 days.
 - The constraints of putting down shelves. Estimated workload, 2 days.
 - The constraints of picking up shelf. Estimated workload, less than 1 day.
 - The constraints of shelf action are bind to the same robot picking it up. Estimated workload, 1 day.

This is a rough breakdown of the tasks and a sketchy timetable for completeness. By estimation, to complete this project will take about a week and a half. This schedule is based on our approach aims to write down test cases during the phases of development and make sure the cases pass before continuing to the next phase. We will also have to consider leaving some of the small working items undone if our progress falls behind. Therefore, if unpredictable situation happens, we will forfeit the development work in

segment 2, if any, by the 5th day week 6. When we will turn our focus on segment 3 and 4, with 6 days left for us to meet the deadline. We will need to account for a 2 to 3 days span to prepare for the project report, the estimated date on halting all unfinished development work and start completing the project report will be on the 4th day of week 7.

References

ASP Challenge Problem: Automated Warehouse Scenario

Gebser, M. and Obermeier, P. (no date) ASP challenge 2019, ASP Challenge Problem: Automated Warehouse Scenario.

CA: University of Potsdam